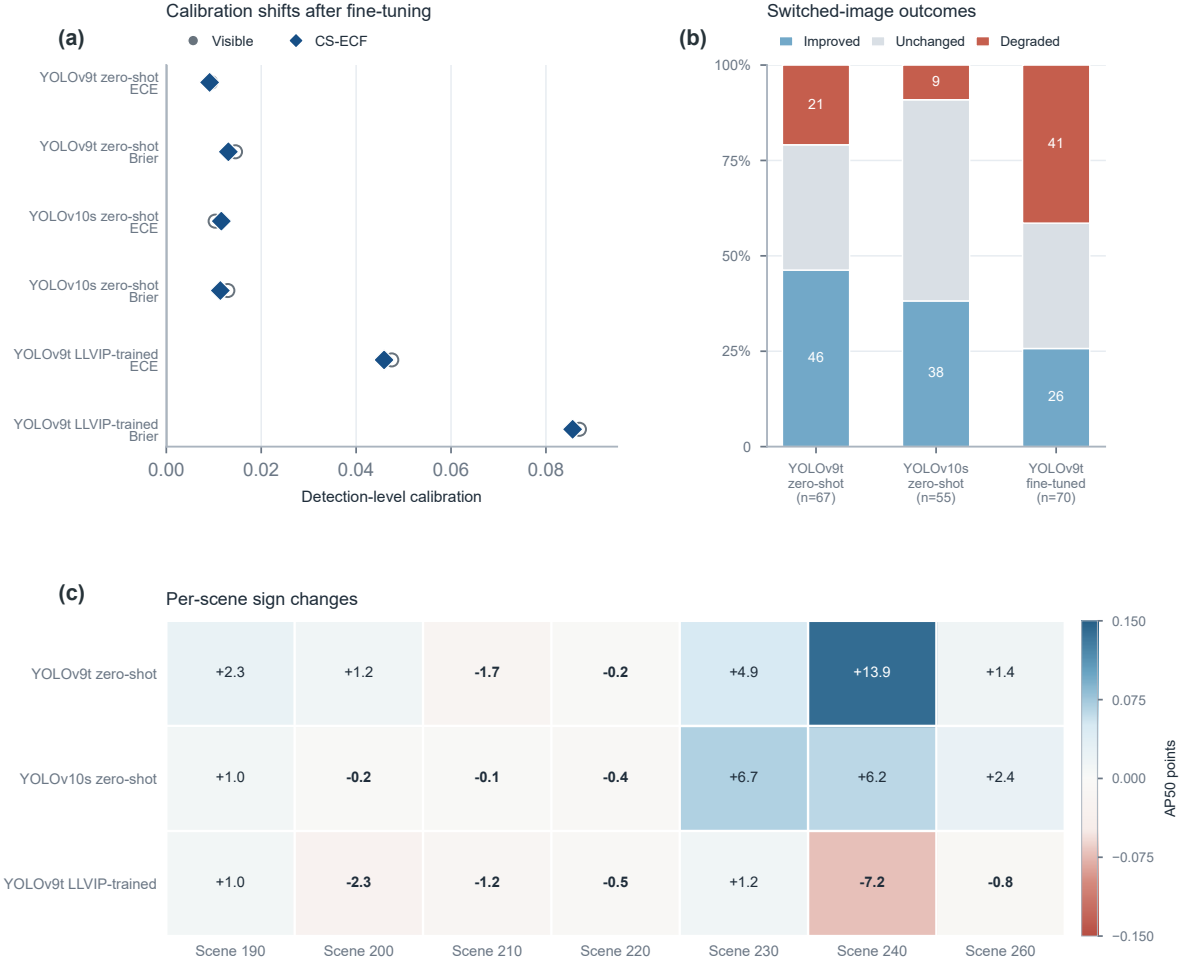

(a)
Calibration shifts after fine-tuning
Visible
CS-ECF
YOLOv9t zero-shot ECE
YOLOv9t zero-shot Brier
YOLOv10s zero-shot ECE
YOLOv10s zero-shot Brier
YOLOv9t LLVIP-trained ECE
YOLOv9t LLVIP-trained Brier
0.00
0.02
0.04
0.06
0.08
Detection-level calibration
(b)
Switched-image outcomes
Improved
Unchanged
Degraded
100%
75%
50%
25%
0
21
46
9
38
41
26
YOLOv9t zero-shot (n=67)
YOLOv10s zero-shot (n=55)
YOLOv9t fine-tuned (n=70)
(c)
Per-scene sign changes
YOLOv9t zero-shot
YOLOv10s zero-shot
YOLOv9t LLVIP-trained
+2.3
+1.2
-1.7
-0.2
+4.9
+13.9
+1.4
+1.0
-0.2
-0.1
-0.4
+6.7
+6.2
+2.4
+1.0
-2.3
-1.2
-0.5
+1.2
-7.2
-0.8
Scene 190
Scene 200
Scene 210
Scene 220
Scene 230
Scene 240
Scene 260
0.150
0.075
0.000
−0.075
−0.150
AP50 points